

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026 — ANSWER KEY (Version 1)

Chemistry — Class 9 (Science Group) | Exam-18 | Total Marks: 31

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table & Periodicity | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

SECTION A — MCQ Answers (12 × 1 = 12 Marks)

1. Branch designing safer products and processes → **B. Green chemistry** 1

Green chemistry designs chemical products and processes that **reduce or eliminate the use and generation of hazardous substances**. Industrial chemistry deals with large-scale manufacture, nuclear chemistry with radioactivity, and astrochemistry with matter in space.

2. Liquid crystals are used in → **B. display screens (LCDs)** 1

Liquid crystals are an intermediate state between liquids and solids: the molecules **flow like a liquid but keep some ordered structure** like a solid. This is what makes them useful in **liquid crystal displays (LCDs)**.

3. Element existing in different physical forms → **B. allotropy** 1

Allotropy is the property of an element to exist in different physical forms in the **same physical state**; those forms are called **allotropes**. Diamond, graphite and buckyballs are the three allotropes of carbon. Isotopy is about atoms of the same element with different mass numbers, which is a different idea altogether.

4. Neutrons in ^{56}Fe ($Z = 26$) → **B. 30** 1

Number of neutrons = **mass number – atomic number** = $56 - 26 = 30$. The atomic number 26 gives the number of protons (and, in a neutral atom, the electrons).

5. Discovery of atomic number in 1913 → **B. Moseley** 1

After **Moseley** discovered the atomic number in **1913**, it was realised that atomic number, not atomic mass, is the correct basis for arranging the elements. This gave the **modern periodic law**.

6. Number of elements in the sixth period → **D. 32** 1

The periodic table has **seven periods**, and the number of elements per period ranges from **2 in period 1 to 32 in period 6**. The first three periods are called **short periods** and the remaining ones **long periods**.

7. Each K atom in K_2O → **B. loses 1 electron** 1

Potassium is in **Group 1** and has **1 valence electron**. It **loses** that one electron to reach a stable noble-gas arrangement, forming K^+ . Oxygen (Group 16) gains 2 electrons to form O^{2-} , so **two** K atoms are needed for every one O atom — hence the formula K_2O .

8. Empirical formula of ethene C_2H_4 → **B. CH_2** 1

The **empirical formula** shows the **simplest whole-number ratio** of atoms. Dividing the subscripts of C_2H_4 by 2 gives **CH_2** .

9. Term "formula mass" is used for → **B. NaCl** 1

NaCl is an **ionic compound**. It does not exist as separate molecules but as a giant lattice of Na^+ and Cl^- ions, so we speak of its **formula mass**, not its molecular mass. CO_2 , H_2O and CH_4 are covalent and do form real molecules.

10. Coefficient of O_2 when balancing $\text{C}_2\text{H}_2 + \text{O}_2$ → **C. 5** 1

The balanced equation is **$2 \text{C}_2\text{H}_2 + 5 \text{O}_2 \rightarrow 4 \text{CO}_2 + 2 \text{H}_2\text{O}$** .

Check — C: $4 = 4$ | H: $4 = 4$ | O: $10 = 8 + 2 = 10$. ✓

11. Oxidation number of Cl in $\text{KClO}_2 \rightarrow \text{B. } +3$ **1**

K = +1, each O = -2, and the compound is neutral:

$$(+1) + \text{Cl} + 2(-2) = 0 \rightarrow 1 + \text{Cl} - 4 = 0 \rightarrow \text{Cl} = +3.$$

12. Enthalpy relationship in an exothermic reaction $\rightarrow \text{B. } H_{\text{products}} < H_{\text{reactants}}$ **1**

In an **exothermic** reaction heat is **released** to the surroundings, so the products end up storing **less** energy than the reactants. Since $\Delta H = H_{\text{products}} - H_{\text{reactants}}$, this makes **ΔH negative**. Options A and D describe an **endothermic** reaction.

SECTION B — Short Question Answers ($3 \times 3 = 9$ Marks)

Q.2 (i) Molar mass of SO_2 and the mass of one molecule. **3**

(a) Molar mass of SO_2 **1**

$$M = (1 \times \text{S}) + (2 \times \text{O}) = (1 \times 32) + (2 \times 16) = 32 + 32$$

$$M = 64 \text{ g/mol}$$

(b) Mass of one molecule **2**

One mole of SO_2 has a mass of 64 g and contains $N_A = 6.02 \times 10^{23}$ molecules.

Mass of one molecule = molar mass $\div N_A$ **1**

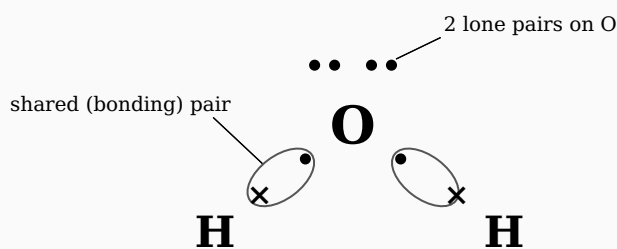
$$= 64 \div (6.02 \times 10^{23})$$

$$= 10.63 \times 10^{-23}$$

$$= 1.06 \times 10^{-22} \text{ g} \quad \textbf{1}$$

Q.2 (ii) Dot-and-cross structure of water, bond type, lone pairs. **3**

Oxygen ($Z = 8$) has the configuration **2, 6**, so it has **6** valence electrons and needs 2 more to complete its octet. Each hydrogen has **1** electron and needs 1 more for its duplet. So oxygen shares **one** electron with each of the **two** hydrogen atoms.



Key: $\bullet$ = electron of oxygen $\times$ = electron of hydrogen

1 correct dot-and-cross diagram (2 shared pairs + 2 lone pairs on O; oxygen shown with 8 electrons and each hydrogen with 2).

Type of bond: a **single covalent bond** between O and each H (formed by **mutual sharing** of electrons). Because oxygen is far more electronegative than hydrogen, the shared pairs lie closer to oxygen, so each bond is a **polar covalent** bond. **1**

Lone pairs on oxygen: **2** (that is, 4 non-bonding electrons). **1**

Q.2 (iii) Oxidation numbers. 3**(a) S in H_2SO_3** (H = +1, O = -2, molecule is neutral) **1**

$$2(+1) + \text{S} + 3(-2) = 0$$

$$2 + \text{S} - 6 = 0$$

$$\text{S} = +4$$

(b) C in Na_2CO_3 (Na = +1, O = -2) **1**

$$2(+1) + \text{C} + 3(-2) = 0$$

$$2 + \text{C} - 6 = 0$$

$$\text{C} = +4$$

(c) I in KIO_4 (K = +1, O = -2) **1**

$$(+1) + \text{I} + 4(-2) = 0$$

$$1 + \text{I} - 8 = 0$$

$$\text{I} = +7$$

SECTION C — Long Question Answers ($2 \times 5 = 10$ Marks)**Q.3 (i) Empirical formula, molecular formula and moles. 5****(a) Empirical formula 3**Take a **100 g** sample, so the percentages become masses in grams.

Element	Mass in 100 g	÷ atomic mass = moles	÷ smallest (2.2)	Simplest ratio
Carbon	26.7 g	$26.7 \div 12 = 2.225$	$2.225 \div 2.2 = 1.01$	1
Hydrogen	2.2 g	$2.2 \div 1 = 2.2$	$2.2 \div 2.2 = 1.00$	1
Oxygen	71.1 g	$71.1 \div 16 = 4.444$	$4.444 \div 2.2 = 2.02$	2

1 converting to moles **1** dividing by the smallest **1** final answer**Empirical formula = CHO_2** **(b) Molecular formula 1**

$$\text{Empirical formula mass} = 12 + 1 + 2(16) = 12 + 1 + 32 = \mathbf{45}$$

$$n = \text{molar mass} \div \text{empirical formula mass} = 90 \div 45 = \mathbf{2}$$

$$\text{Molecular formula} = (\text{CHO}_2)_2 = \mathbf{C_2H_2O_4}$$

(c) Moles in 18 g 1

$$n = \text{mass} \div \text{molar mass}$$

$$n = 18 \div 90$$

$$\mathbf{n = 0.2 \text{ mol}}$$

Q.3 (ii) Relative atomic mass of silver; sub-atomic particles; nitrogen configuration. **5**

(a) Relative atomic mass of silver **2**

Relative atomic mass is the **weighted average** of the masses of all the isotopes:

$$\text{R.A.M.} = [(\text{mass}_1 \times \%_1) + (\text{mass}_2 \times \%_2)] \div 100 \quad \mathbf{1}$$

$$\text{R.A.M.} = [(107 \times 52) + (109 \times 48)] \div 100$$

$$\text{R.A.M.} = (5564 + 5232) \div 100$$

$$\text{R.A.M.} = 10796 \div 100$$

$$\text{R.A.M. of silver} = \mathbf{107.96} \quad \mathbf{1}$$

(This is why the periodic table shows silver as about 108, not a whole number.)

(b) Sub-atomic particles in one atom of ^{109}Ag **1**

Particle	Number	How it is found
Protons	47	= atomic number Z
Electrons	47	= number of protons (the atom is neutral)
Neutrons	62	= mass number – atomic number = $109 - 47$

(c) Nitrogen, atomic number 7 **2**

Electronic configuration: $1s^2 2s^2 2p^3$ (shell notation: 2, 5) **1**

Group: the valence shell ($n = 2$) holds $2 + 3 = \mathbf{5}$ **electrons**, so nitrogen belongs to **Group 15 (VA)**. The number of valence electrons decides the group. $\frac{1}{2}$

Period: the highest occupied shell is $n = \mathbf{2}$, that is, the atom has **two** occupied shells, so nitrogen lies in **Period 2**. The number of occupied shells decides the period. $\frac{1}{2}$

— End of Answer Key —